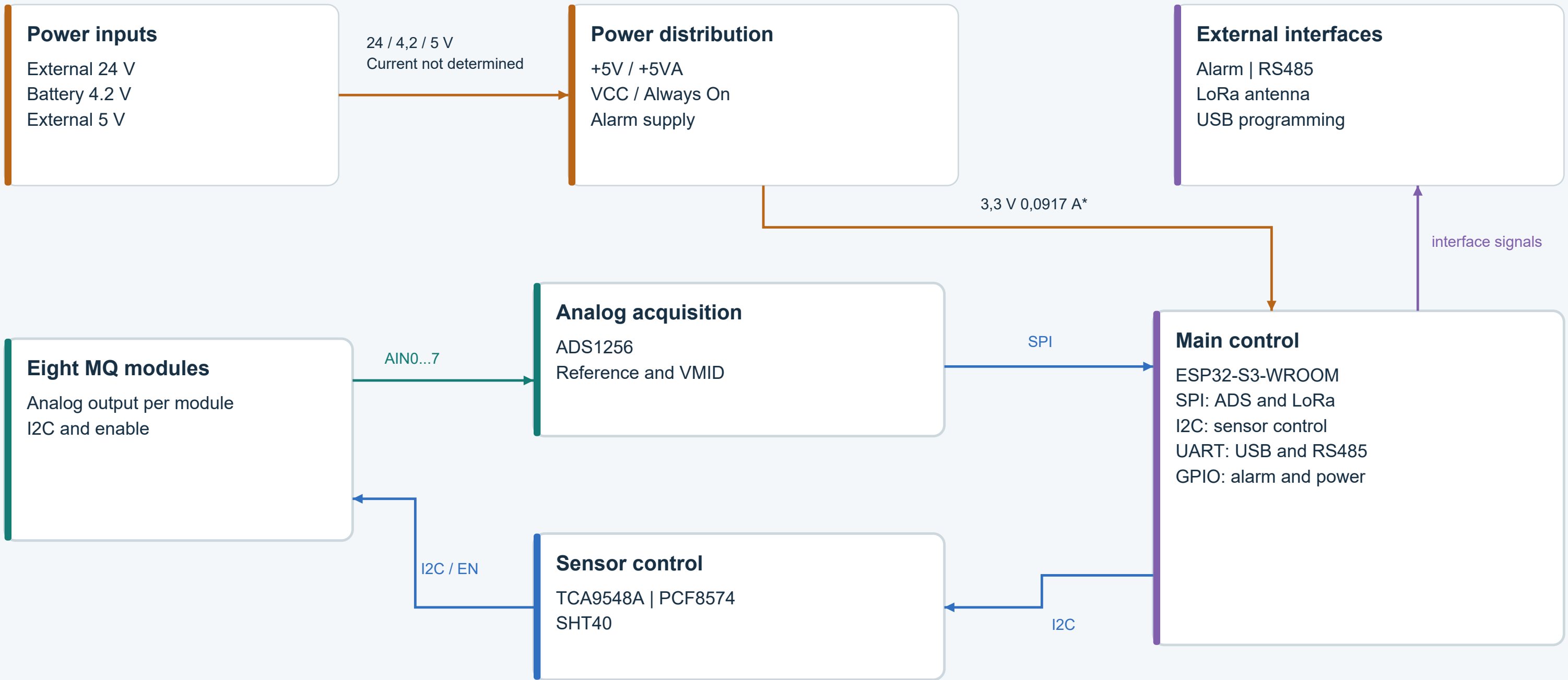


# 01 System architecture

Power, acquisition, control, and communications. Power values are calculated reference cases; see pages 10–17.

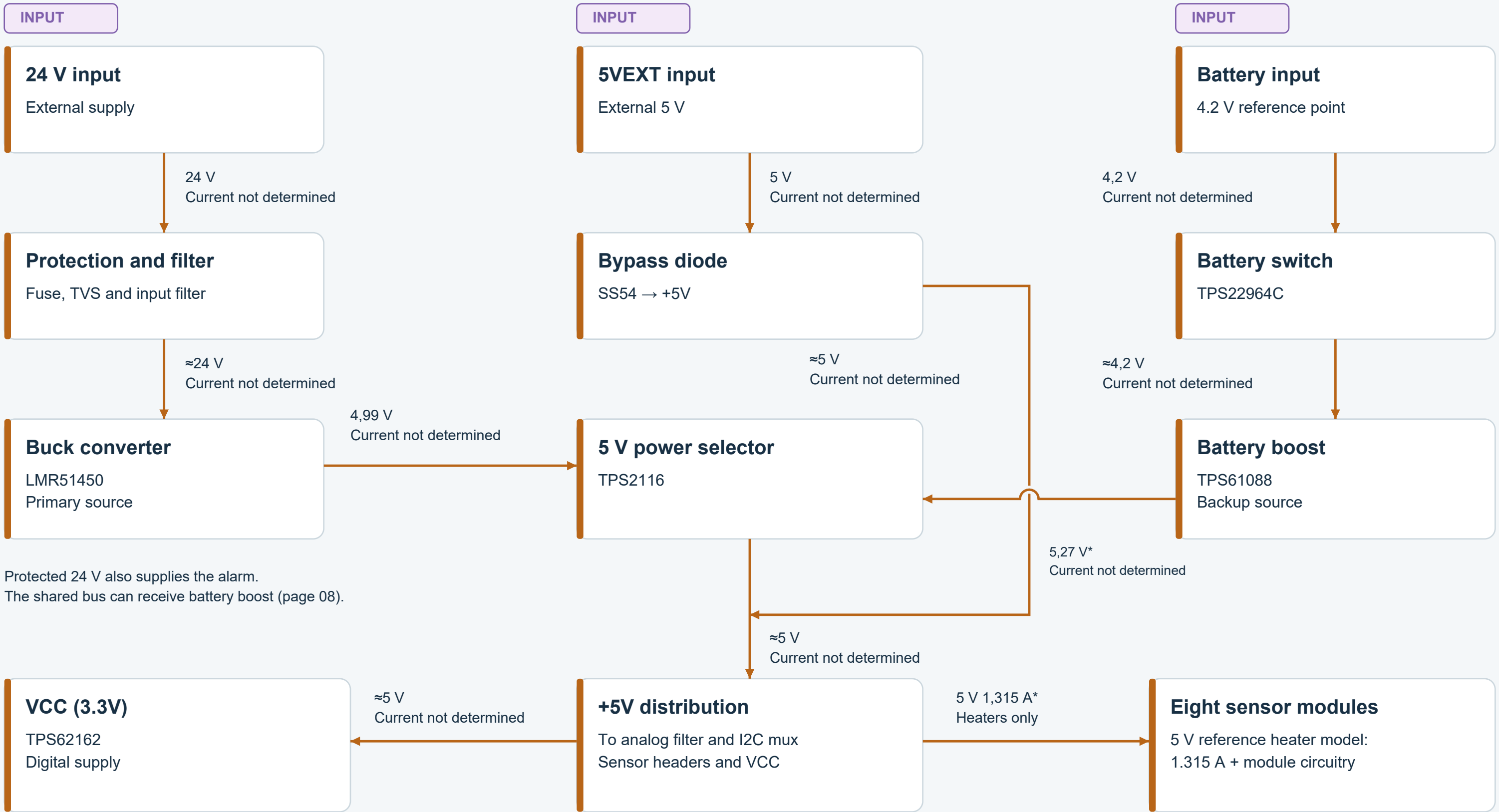


## Reading the power labels

Brown = power. \* = stated reference case, not a measured total. Approximate voltages include source/path variation. Unknown current is written in plain words. Calculation symbols and conditions are kept in the appendix.

# 02 Main power distribution

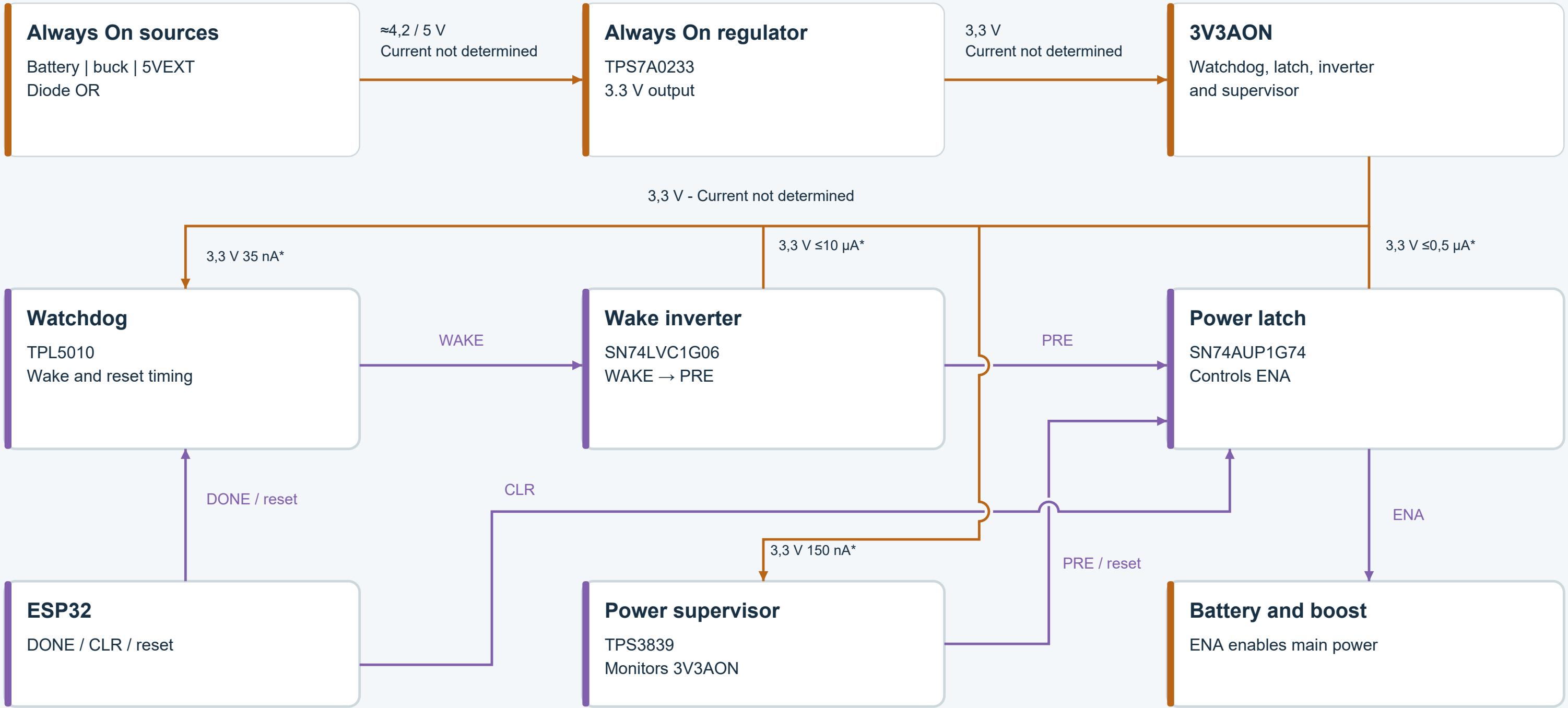
24 V at left, battery at right, source selector at centre. The external 5 V diode bypasses the selector.



\*Boost: 5,27 V PWM / 5,30 V PFM before path losses. Heater current: reference case at exactly 5 V; module ICs are additional.

### 03 Always On, watchdog, and latch

Purpose: retain supervision and wake control while the main switched circuitry is disabled.

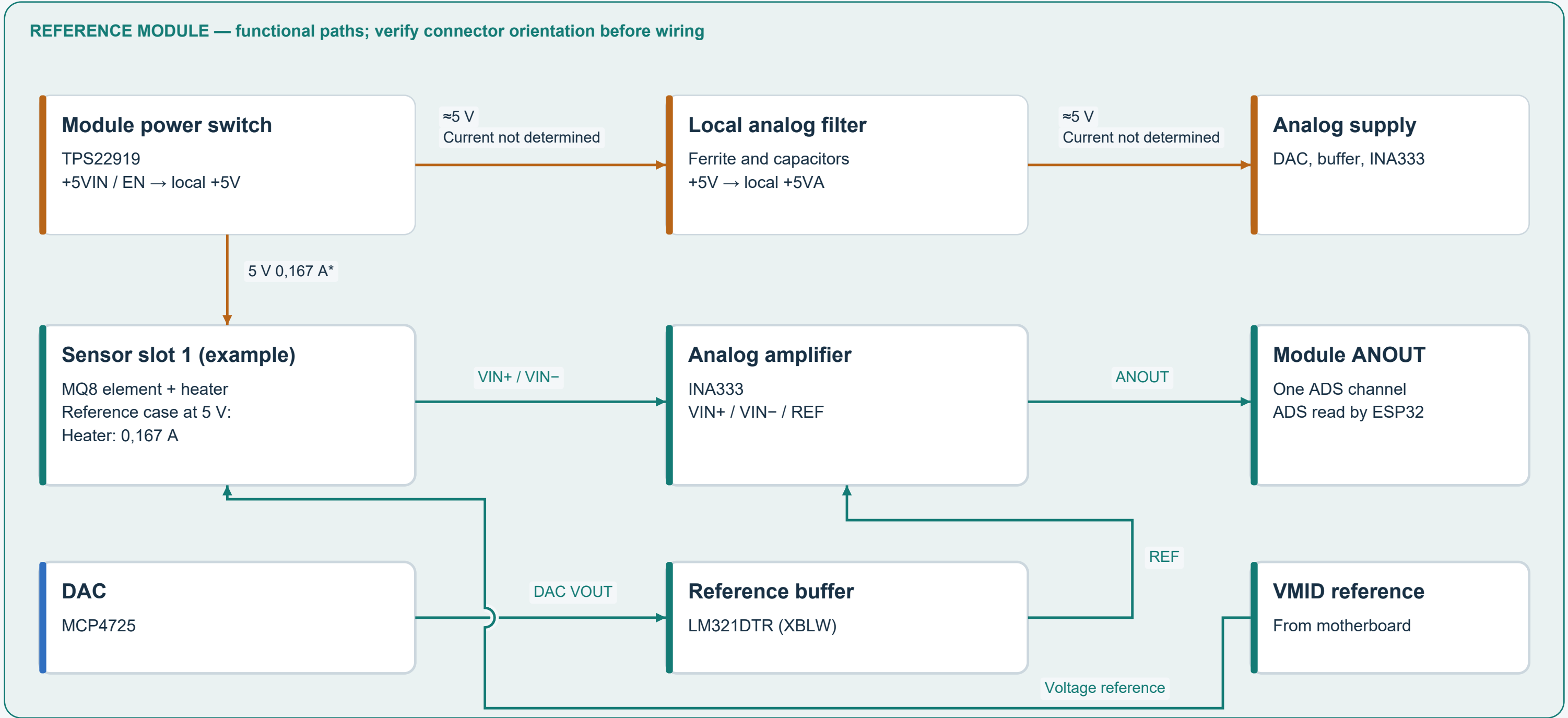


#### AON budget

\*Static, unloaded reference cases; conditions on page 13. Total current also includes pull-ups, switching activity and regulator losses.

# 04 Sensor module: functional path

One example of eight modules. Local analog supply is separate from the motherboard analog supply.

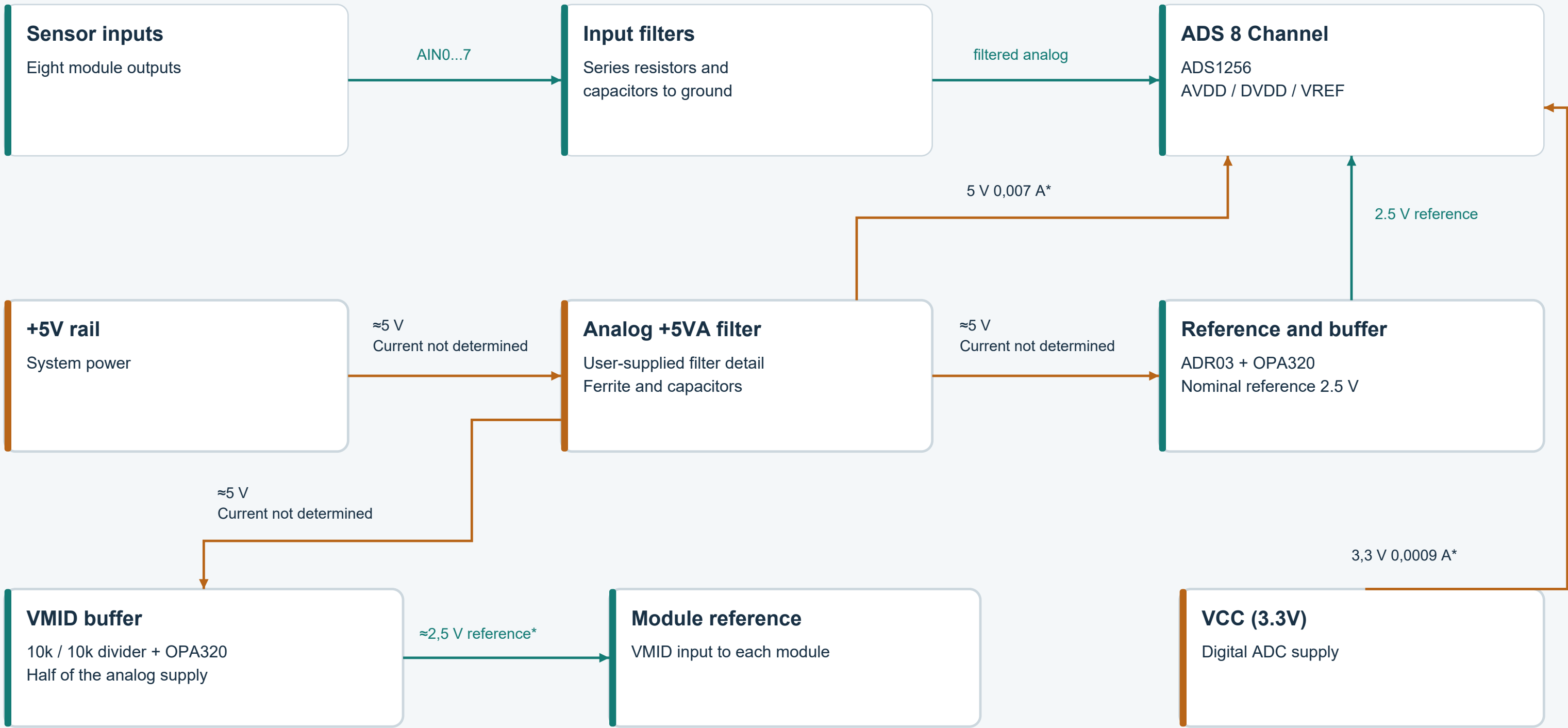


## Module input current

\*Heater reference model. Module current includes heater, analog circuitry and switch. Sensing current comes from the motherboard reference.

# 05 Analog acquisition and voltage reference

ADS1256 multiplexes eight analog inputs. Channels are converted sequentially.

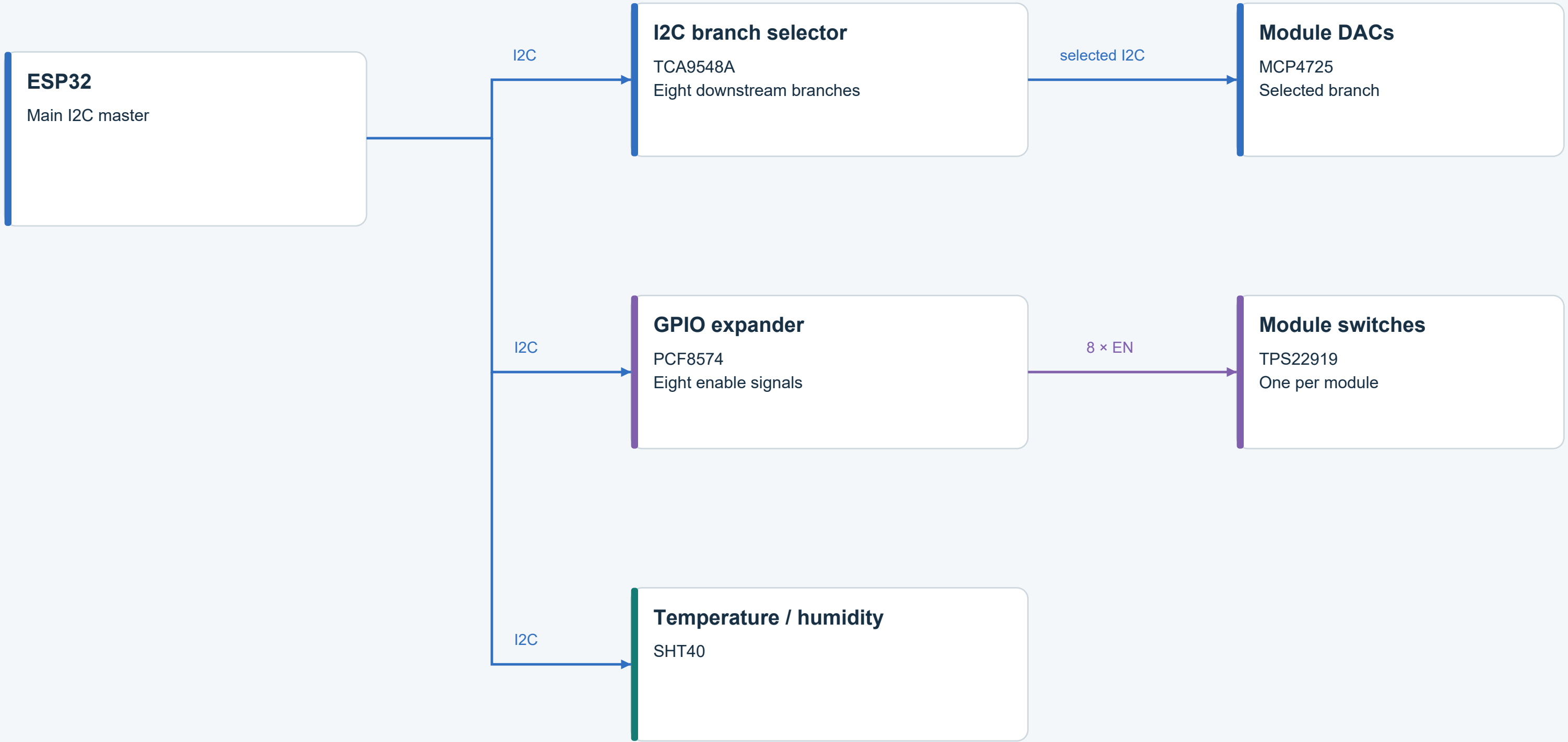


\*Reference values: analog supply 5 V, digital supply 3,3 V. Midpoint voltage changes with supply and output load.

At the datasheet reference point: ADS analog 7 mA + digital 0.9 mA typical = 37.97 mW. Conditions: page 13.

# 06 I2C and sensor control

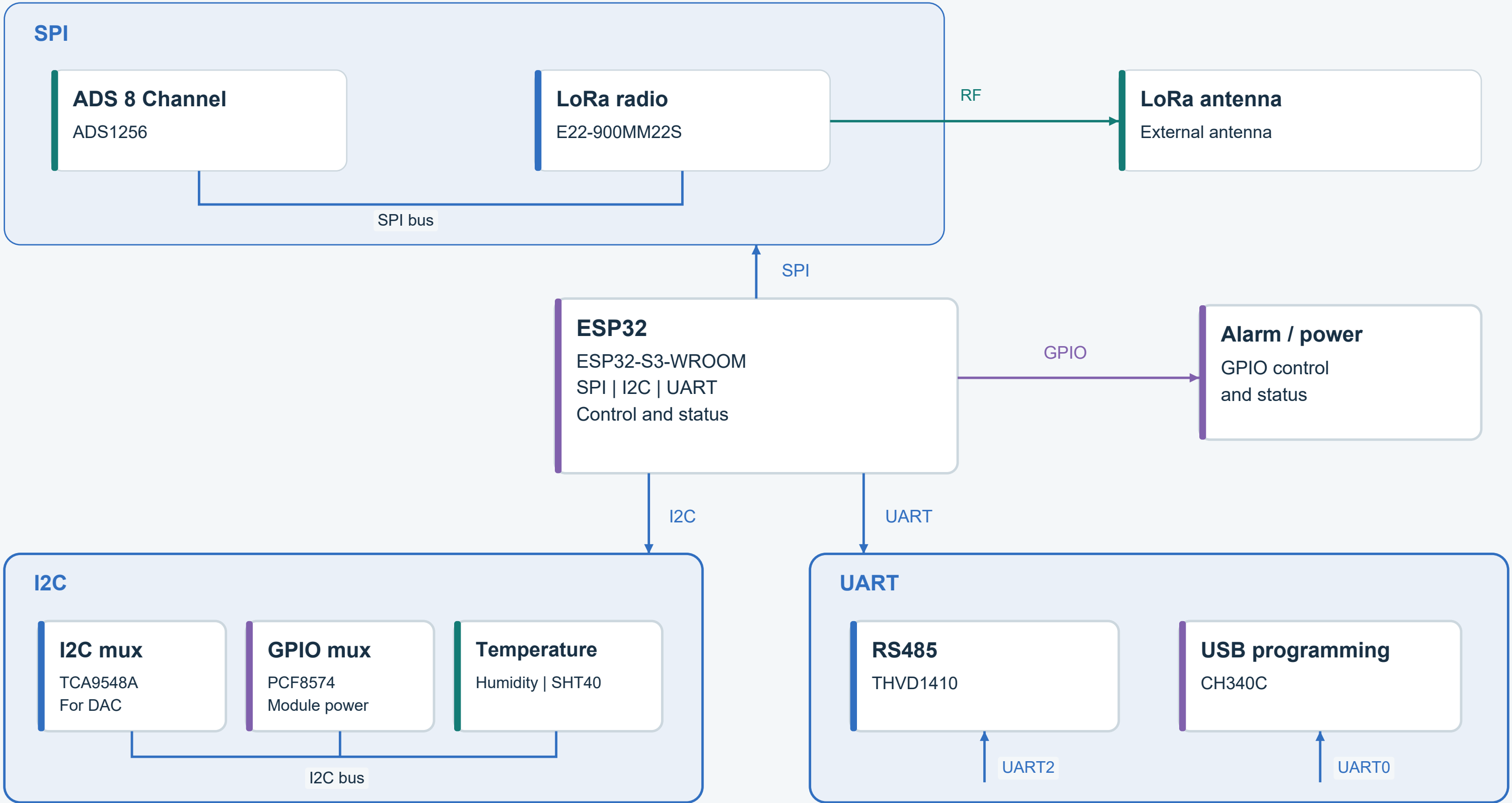
The ESP32 addresses three devices on the main I2C bus. Only the sensor DAC branches pass through the I2C multiplexer.



Supply: TCA9548A = +5V; PCF8574 and SHT40 = VCC (3.3V). Current references: pages 12–14.

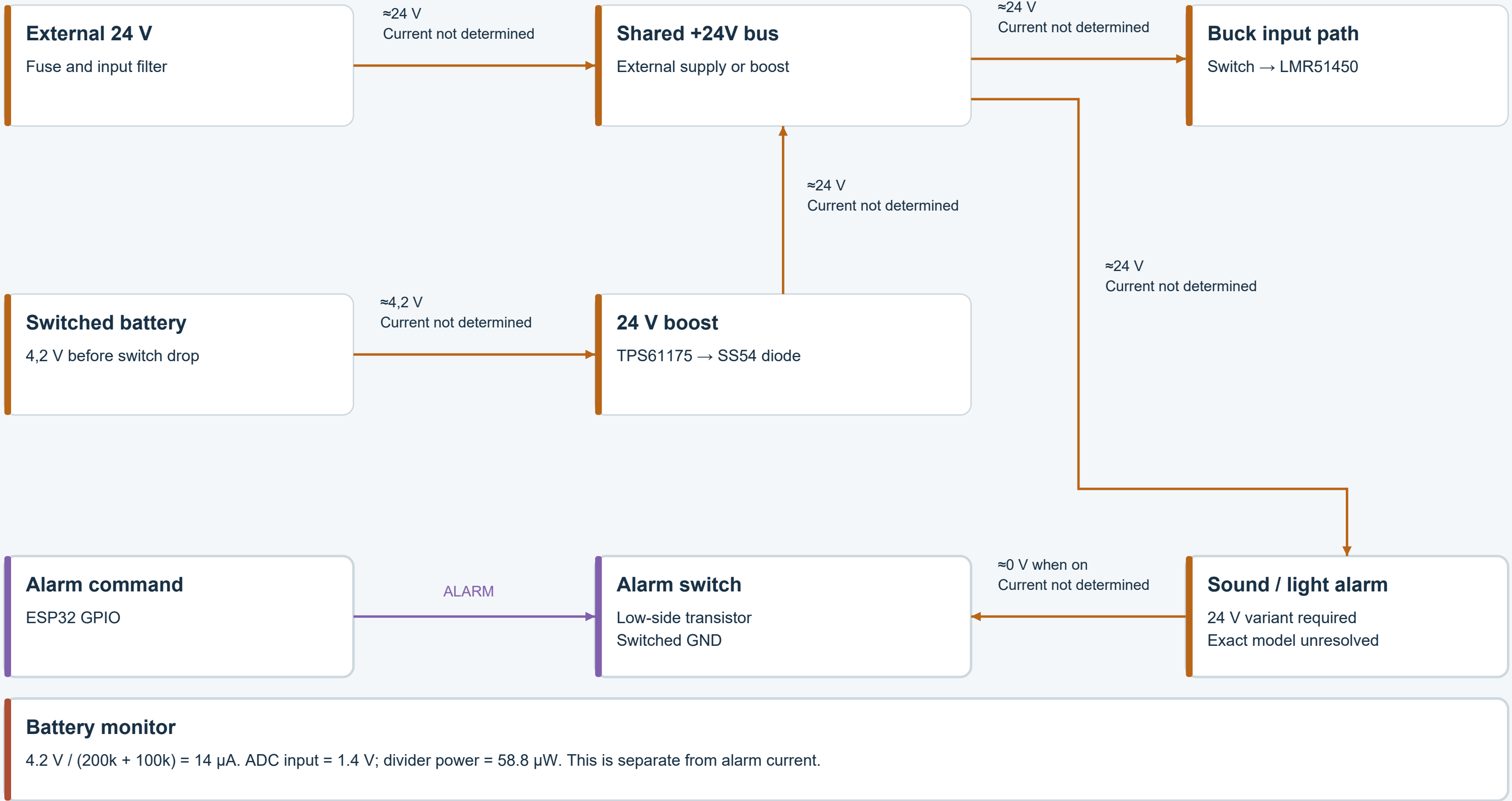
# 07 ESP32 communications and connections

One link to each bus group. Branches show shared SPI/I2C and separate UART ports.



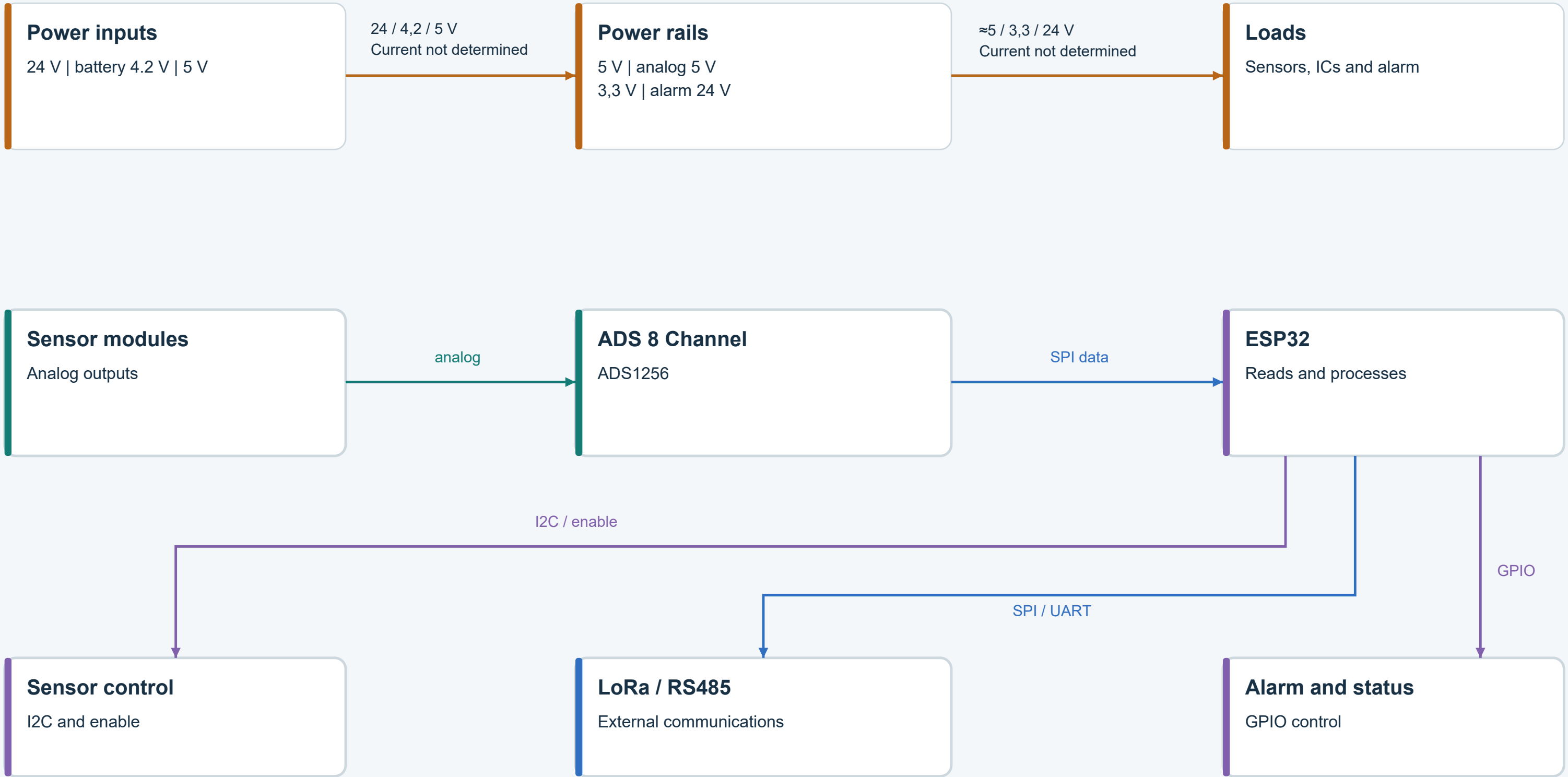
# 08 Shared 24 V rail, alarm, and battery

The +24V bus feeds both the alarm and buck converter. Battery boost demand can include both loads.



# 09 Energy, data, and control flow

Power feeds devices. Sensor data travels through ADS to ESP32, then to communications interfaces.



# 10 Voltage calculations

Nominal resistor values from the schematic. These values exclude tolerance, ripple and load-dependent path drops.

| Rail / device            | Calculation from schematic        | Result     | Source    |
|--------------------------|-----------------------------------|------------|-----------|
| 5VBUCK / LMR51450        | $0.8 \times (1 + 100k / 19.1k)$   | 4.98848 V  | S01       |
| 5VBOOST / TPS61088, PWM  | $1.204 \times (1 + 189k / 56k)$   | 5.26750 V  | S02       |
| 5VBOOST / TPS61088, PFM  | $1.212 \times (1 + 189k / 56k)$   | 5.30250 V  | S02       |
| Alarm boost, before SS54 | $1.229 \times (1 + 187k / 10k)$   | 24.21130 V | S03       |
| VCC / TPS62162           | Fixed output option               | 3.3 V      | S04       |
| 3V3AON / TPS7A0233       | Fixed output option               | 3.3 V      | S06       |
| VMID, unloaded           | $VA \times 10k / (10k + 10k)$     | $VA / 2$   | Schematic |
| Battery monitor at 4.2 V | $4.2 \times 100k / (200k + 100k)$ | 1.4 V      | Schematic |

### Actual rail voltage

V5 = selected buck/boost voltage minus selector drop, or 5VEXT minus diode drop.  
VA = V5 minus analog-filter DC drop.  
VA24 = external or boosted source minus path drops.

### Source compatibility needs verification

The backup set point is above 5.25 V (ADS AVDD limit) and 5.1 V (the reference MQ heater upper limit).  
A diode/filter drop is load-dependent; it does not prove the output is within range.

# 11 Eight MQ heaters: calculated reference

Configuration identifies sensor types; fitted manufacturers are unconfirmed. Sources below are explicit reference vendors.

| Slot / type | Reference | RH ( $\Omega$ ) | I at 5 V (mA) | P at 5 V (W) | I at min RH (mA) | Ref |
|-------------|-----------|-----------------|---------------|--------------|------------------|-----|
| 1 MQ8       | Winsen    | 30 $\pm$ 3      | 166.67        | 0.8333       | 185.19           | S20 |
| 2 MQ135     | Winsen    | 30 $\pm$ 3      | 166.67        | 0.8333       | 185.19           | S21 |
| 3 MQ3       | Hanwei    | 33 $\pm$ 1.65   | 151.52        | 0.7576       | 159.49           | S22 |
| 4 MQ5       | Winsen    | 30 $\pm$ 3      | 166.67        | 0.8333       | 185.19           | S23 |
| 5 MQ4       | Winsen    | 28 $\pm$ 3      | 178.57        | 0.8929       | 200.00           | S24 |
| 6 MQ7       | Hanwei    | 33 $\pm$ 1.65   | 151.52        | 0.7576       | 159.49           | S25 |
| 7 MQ6       | Winsen    | 30 $\pm$ 3      | 166.67        | 0.8333       | 185.19           | S26 |
| 8 MQ2       | Winsen    | 30 $\pm$ 3      | 166.67        | 0.8333       | 185.19           | S27 |

### All eight enabled at exactly 5.000 V

$I = \Sigma(5 / RH) = 1.31494 \text{ A}$   
 $P = \Sigma(25 / RH) = 6.57468 \text{ W}$   
At minimum RH: 1.44491 A (same 5 V).

### Meaning of these numbers

Room-temperature resistance calculation only.  
Not a steady-state heater maximum or measured load.  
Add sensing paths, analog ICs and conversion losses.

MQ7: the Hanwei reference specifies 5 V for 60 s and 1.4 V for 90 s. The constant-resistance cycle gives 0.33867 W; that cycle is not present in the supplied fixed-5V module reference. Do not use the cycle average as this board’s continuous-5V load.

# 12 Digital IC load reference cases

P = Vtest × I. Vtest = 3.3 V unless stated. Do not add mutually exclusive operating modes.

| Device / condition                   | Current          | Calculated power       | Basis                               |
|--------------------------------------|------------------|------------------------|-------------------------------------|
| ESP32, 240 MHz dual-core data access | 91.7 mA typ      | 302.61 mW              | RF off; peripheral clocks off [S10] |
| ESP32, Wi-Fi 802.11b TX, 20.5 dBm    | 355 mA peak      | 1171.50 mW             | Separate RF-on condition [S10]      |
| LoRa, maximum TX operating mode      | 119 mA typ       | 392.70 mW              | E22-900MM22S [S11]                  |
| LoRa, receive                        | 6.8 mA typ       | 22.44 mW               | RX only [S11]                       |
| LoRa, sleep                          | 2 µA typ         | 6.60 µW                | Sleep only [S11]                    |
| SHT40, measurement                   | 320 / 500 µA     | 1.056 / 1.650 mW       | Typ / max; heater off [S12]         |
| SHT40, highest heater setting        | 60 / 100 mA      | 198 / 330 mW           | Typ / max; during heating [S12]     |
| THVD1410, receiver only              | 0.700 / 0.960 mA | 2.310 / 3.168 mW       | Typ / max, unloaded [S15]           |
| THVD1410, driver only                | 2.0 / 2.5 mA     | 6.60 / 8.25 mW         | Typ / max; add line load [S15]      |
| PCF8574, NXP reference at 6 V        | 40 / 100 µA      | 240 / 600 µW at 6 V    | 100 kHz, unloaded [S14]             |
| CH340C, active at 3.3 V              | 4 / 12 mA        | 13.2 / 39.6 mW         | 25°C; excludes USB pins [S32]       |
| TCA9548A, reference at 5.5 V         | 9 / 30 µA        | 49.5 / 165 µW at 5.5 V | 100 kHz, unloaded [S13]             |

PCF8574 vendor is unconfirmed: the NXP 6 V case is a reference, not a verified 3.3 V board value. TCA9548A is on +5V; its table above uses the specified 5.5 V test point. Its 400 kHz reference rises to 50 / 80 µA. Do not silently use either test voltage as the actual rail.

ESP32 memory/I/O loads and duty cycle remain additional conditions. VCC also supplies ADS DVDD, pull-ups and LED. CH340C USB suspend is 40 µA typ / 150 µA max at 3.3 V (132 / 495 µW).

# 13 Analog, AON, and module IC references

No-load IC figures do not include pull-ups, output loads, resistor networks or conversion losses.

| Device / condition                     | Current reference           | Power / calculation              | Source    |
|----------------------------------------|-----------------------------|----------------------------------|-----------|
| ADS AVDD at 5 V, PGA1, buffer off      | 7 / 10 mA                   | 35 / 50 mW (typ / max)           | S07       |
| ADS DVDD at 3.3 V, CLKOUT off          | 0.9 / 2 mA                  | 2.97 / 6.6 mW (typ / max)        | S07       |
| OPA320, each, no load at 5.5 V         | 1.5 / 1.75 mA               | 8.25 / 9.625 mW at 5.5 V, 25°C   | S08       |
| ADR03, no load at 5 V                  | 0.65 / 1 mA                 | 3.25 / 5 mW (typ / max)          | S09       |
| VMID divider at VA = 5 V               | 5 / 20k = 250 µA            | 1.25 mW                          | Schematic |
| TPL5010, normal operation              | 35 / 50 nA                  | 0.1155 / 0.165 µW at 3.3 V       | S16       |
| SN74AUP1G74, static, up to 85°C        | 0.5 µA max                  | 1.65 µW at 3.3 V                 | S17       |
| SN74LVC1G06, static                    | 10 µA max                   | 33 µW at 3.3 V                   | S18       |
| TPS3839, no output load                | 150 / 500 nA                | 0.495 / 1.65 µW at 3.3 V         | S19       |
| Module: INA333 + MCP4725 + XBLW buffer | 50 + 210 + 100 = 360 µA typ | IC benchmark only; + other loads | S28–S31   |
| Module switch TPS22919                 | 8 µA typ at 3.6 V           | 28.8 µW at 3.6 V; + I² × RON     | S30       |

ADS figures use the datasheet’s 7.68 MHz clock. The schematic has an 8 MHz crystal: these are reference cases, not guaranteed board maxima. Higher PGA/buffer settings increase analog current. Sum at the stated low-power case: 37.97 mW typ, 56.60 mW from table maxima.

Module benchmark conditions: INA333 at midsupply/no output load; MCP4725 code 000h, digital inputs grounded/no load; XBLW LM321DTR at 5 V/no load. The switch’s 3.6 V figure is not a measured 5 V total. VMID and VREF output currents must be added to their buffers.

# 14 Rail currents and complete budget equations

All I symbols are in amperes, P in watts.  $\eta$  is actual efficiency at the operating point; no efficiency is invented.

| Power path / symbol           | Current / power equation or included load                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------|
| VCC: I33; P33 = 3.3 × I33     | IESP + ILORA + IADSD + ISHT + IPCF + IRS485 + IUSB + IPULL/LED                            |
| VCC converter input: I33IN    | P33 / (V5 × $\eta$ 33). $\eta$ 33 includes converter losses; do not add IQ twice.         |
| Module: IM; all modules: I8   | IM = IH + IMA + IQSW; I8 = sum of eight IM, including off-state leakage.                  |
| Local module analog: IMA      | IDAC + IINA + IBUFFER + local passive/output loads. VMID-sourced ISENSE is excluded.      |
| Motherboard analog: IANA      | IADSA + IREFSUP + IMIDSUP; add capacitor charging/leakage for transients.                 |
| IREFSUP / IMIDSUP             | ADR03 + reference OPA/load / VMID OPA + sensing/reference loads + VA / 20k.               |
| Main rail: I5; P5 = V5 × I5   | I8 + IANA + I33IN + ITCA + I5PULL + leakage; no upstream regulator double count.          |
| Selector paths: IK / IX / IE5 | Source share of I5 + applicable bias/leakage current. Conduction loss is a power term.    |
| Battery boost input: IB5      | Boost output power / (VB × $\eta$ B5); output includes IX and feedback/control loads.     |
| Shared bus: I24EXT + I24BOOST | IAL + IKIN + I24PAR. Alarm branch PALBR = VA24 × IAL, including switch loss.              |
| 24 V boost input: IB24        | P24OUT / (VB × $\eta$ B24); include I24BOOST, diode loss and feedback load.               |
| Switched battery: IBSW        | IB5 + IB24 + other switched-VBAT leakage.                                                 |
| Battery source: IB at 4.2 V   | IBSW + switch ground current + 14 $\mu$ A monitor + battery share of IAIN.                |
| External 24 V input: I24      | I24EXT + input leakage; IKIN = PK / (V24P × $\eta$ K). Series loss changes voltage/power. |
| External 5 V input: IE        | IE5 + external-5V share of IAIN; diode losses set the downstream voltage.                 |

IA = IWD + IINV + ILATCH + ISUP + AON pull-ups/switching; IAIN = IA + LDO ground current IG. PK includes selector demand, buck-fed AON, feedback and PR1 divider loads. I24PAR includes other shared-rail leakage. Set disabled-converter terms to their off-state leakage, not an assumed active load.

VM = module switched +5V; VMA = local analog rail; VA = motherboard analog rail. VD = diode drop. VSW(on) = low-side switch drop; alarm load power = (VA24 – VSW) × IAL.

# 15 Engineering notes and unresolved quantities

Schematic-based calculation boundaries. These are document corrections and findings, not hardware modifications.

## 4.2 V battery: meaningful lower-bound example

At the 5.000 V heater resistance reference:  
6.57468 W / 4.2 V = 1.56540 A.  
This is the ideal source current for that heater case only.  
Actual current includes electronics, alarm and losses.  
No battery runtime is claimed without capacity.

## Alarm cannot yet have a numeric load

BYS / YS-01 / YS-02 is not one identified part.  
The schematic supplies a nominal 24 V alarm.  
Ex marking and sound level do not determine watts.  
Keep IAL symbolic until manufacturer, 24 V variant,  
operating current and inrush are identified.

## Small paths that must remain in the budget

Buck FB: 4.98848 / 119.1k = 41.885 µA.  
Selector PR1: 4.98848 / 20.99k = 237.660 µA.  
Boost FB: 5.26750 / 245k = 21.500 µA.  
Alarm FB: 24.21130 / 197k = 122.900 µA.  
A low 10k pull-up draws Vrail / 10k while low.

## Limits are not consumption

TPS62162: 1 A output rating, not a 1 A load.  
TPS2116: 2.5 A rating, not normal supply current.  
TPS61175: 3 A switch rating, not 3 A at 24 V.  
Check thermal, inductor, switch and transient limits  
with the complete operating-point budget.

Passive-load handling: capacitor charging requires  $I = C \times dV/dt$ . Ferrite “600 Ω” is an AC impedance label, not DC resistance. The module’s 1k QOD resistor is a discharge path, not a continuous 5 mA on-state load. Reference/filter resistors do not automatically draw  $V/R$  when their far end is high impedance.

Provenance: motherboard connections come from the original ZIP and PCB net map; the +5V-to-motherboard-+5VA filter is from the user’s later 5VA.png detail. The reference module is a separate image. Sensor names come from BoardPinsGLD2.h. Original design files and firmware were not changed.

Source interaction: the external +24V filter has no series reverse-block diode; boost operation and backfeed depend on source/control states. USB VBUS also reaches the AON diode OR. These paths prevent an unconditional single total-current figure.

# 16 Source references — power and digital

Manufacturer documents and original manufacturer datasheets mirrored by distributors. Accessed 22 September 2026.

|                                                                                                                                                                                                                    |                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>S01</b>                                                                                                                                                                                                         | <b>LMR51450 — TI electrical characteristics: 0.8 V feedback</b>        |
| <a href="http://www.ti.com/lit/ds/symlink/lmr51450.pdf">www.ti.com/lit/ds/symlink/lmr51450.pdf</a>                                                                                                                 |                                                                        |
| <b>S02</b>                                                                                                                                                                                                         | <b>TPS61088 — TI electrical characteristics: PWM/PFM feedback</b>      |
| <a href="http://www.ti.com/lit/ds/symlink/tps61088.pdf">www.ti.com/lit/ds/symlink/tps61088.pdf</a>                                                                                                                 |                                                                        |
| <b>S03</b>                                                                                                                                                                                                         | <b>TPS61175 — TI electrical characteristics: 1.229 V feedback</b>      |
| <a href="http://www.ti.com/lit/ds/symlink/tps61175.pdf">www.ti.com/lit/ds/symlink/tps61175.pdf</a>                                                                                                                 |                                                                        |
| <b>S04</b>                                                                                                                                                                                                         | <b>TPS62162 — TI fixed 3.3 V option; converter rating</b>              |
| <a href="http://www.ti.com/lit/ds/symlink/tps62162.pdf">www.ti.com/lit/ds/symlink/tps62162.pdf</a>                                                                                                                 |                                                                        |
| <b>S05</b>                                                                                                                                                                                                         | <b>TPS2116 — TI power multiplexer; rating is not load current</b>      |
| <a href="http://www.ti.com/lit/ds/symlink/tps2116.pdf">www.ti.com/lit/ds/symlink/tps2116.pdf</a>                                                                                                                   |                                                                        |
| <b>S06</b>                                                                                                                                                                                                         | <b>TPS7A02 — TI fixed 3.3 V option; ground current depends on load</b> |
| <a href="http://www.ti.com/lit/ds/symlink/tps7a02.pdf">www.ti.com/lit/ds/symlink/tps7a02.pdf</a>                                                                                                                   |                                                                        |
| <b>S07</b>                                                                                                                                                                                                         | <b>ADS1256 — TI electrical characteristics, 7.68 MHz clock</b>         |
| <a href="http://www.ti.com/lit/ds/symlink/ads1256.pdf">www.ti.com/lit/ds/symlink/ads1256.pdf</a>                                                                                                                   |                                                                        |
| <b>S08</b>                                                                                                                                                                                                         | <b>OPA320 — TI supply-current table, no output load</b>                |
| <a href="http://www.ti.com/lit/ds/symlink/opa320.pdf">www.ti.com/lit/ds/symlink/opa320.pdf</a>                                                                                                                     |                                                                        |
| <b>S09</b>                                                                                                                                                                                                         | <b>ADR03 — Analog Devices Rev. S, Table 4</b>                          |
| <a href="http://www.analog.com/media/en/technical-documentation/data-sheets/ADR01_02_03_06.pdf">www.analog.com/media/en/technical-documentation/data-sheets/ADR01_02_03_06.pdf</a>                                 |                                                                        |
| <b>S10</b>                                                                                                                                                                                                         | <b>ESP32-S3-WROOM-1 — Espressif v1.8, Tables 6-4 and 6-6</b>           |
| <a href="http://www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf">www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf</a> |                                                                        |
| <b>S11</b>                                                                                                                                                                                                         | <b>E22-900MM22S — Ebyte electrical-parameter table</b>                 |
| <a href="http://www.cdebyte.com/products/E22-900MM22S/1">www.cdebyte.com/products/E22-900MM22S/1</a>                                                                                                               |                                                                        |
| <b>S12</b>                                                                                                                                                                                                         | <b>SHT40 — Sensirion SHT4x current and heater tables</b>               |
| <a href="http://sensirion.com/resource/datasheet/sht4x">sensirion.com/resource/datasheet/sht4x</a>                                                                                                                 |                                                                        |
| <b>S13</b>                                                                                                                                                                                                         | <b>TCA9548A — TI electrical characteristics, 100/400 kHz</b>           |
| <a href="http://www.ti.com/lit/ds/symlink/tca9548a.pdf">www.ti.com/lit/ds/symlink/tca9548a.pdf</a>                                                                                                                 |                                                                        |
| <b>S14</b>                                                                                                                                                                                                         | <b>PCF8574 — NXP reference only; fitted vendor not established</b>     |
| <a href="http://www.nxp.com/docs/en/data-sheet/PCF8574_PCF8574A.pdf">www.nxp.com/docs/en/data-sheet/PCF8574_PCF8574A.pdf</a>                                                                                       |                                                                        |
| <b>S15</b>                                                                                                                                                                                                         | <b>THVD1410 — TI quiescent current; excludes bus load</b>              |
| <a href="http://www.ti.com/lit/ds/symlink/thvd1410.pdf">www.ti.com/lit/ds/symlink/thvd1410.pdf</a>                                                                                                                 |                                                                        |
| <b>S16</b>                                                                                                                                                                                                         | <b>TPL5010 — TI supply current and timing conversion</b>               |
| <a href="http://www.ti.com/lit/ds/symlink/tp15010.pdf">www.ti.com/lit/ds/symlink/tp15010.pdf</a>                                                                                                                   |                                                                        |

Full URLs and calculation inputs: power-sources.json, power-calculations.json and power\_budget.py in the diagram package.

# 17 Source references — sensors and analog

Manufacturer documents and original manufacturer datasheets mirrored by distributors. Accessed 22 September 2026.

**S17 SN74AUP1G74 — TI static ICC at valid logic levels**

[www.ti.com/lit/ds/symlink/sn74aup1g74.pdf](http://www.ti.com/lit/ds/symlink/sn74aup1g74.pdf)

**S18 SN74LVC1G06 — TI static ICC; excludes pull-up load**

[www.ti.com/lit/ds/symlink/sn74lvc1g06.pdf](http://www.ti.com/lit/ds/symlink/sn74lvc1g06.pdf)

**S19 TPS3839 — TI supply current, output unloaded**

[www.ti.com/lit/ds/symlink/tps3839.pdf](http://www.ti.com/lit/ds/symlink/tps3839.pdf)

**S20 MQ8 — Winsen MQ-8 heater resistance at room temperature**

[www.winsen-sensor.com/product/mq-8.html](http://www.winsen-sensor.com/product/mq-8.html)

**S21 MQ135 — Winsen MQ135 heater resistance at room temperature**

[www.winsen-sensor.com/manual/mq135.html](http://www.winsen-sensor.com/manual/mq135.html)

**S22 MQ3 — Hanwei original datasheet, hosted by Olimex**

[www.olimex.com/Products/Components/Sensors/Gas/SNS-MQ3/resources/SNS-MQ3.pdf](http://www.olimex.com/Products/Components/Sensors/Gas/SNS-MQ3/resources/SNS-MQ3.pdf)

**S23 MQ5 — Winsen MQ-5 heater resistance at room temperature**

[www.winsen-sensor.com/product/mq-5.html](http://www.winsen-sensor.com/product/mq-5.html)

**S24 MQ4 — Winsen MQ-4 heater resistance at room temperature**

[www.winsen-sensor.com/product/mq-4.html](http://www.winsen-sensor.com/product/mq-4.html)

**S25 MQ7 — Hanwei original datasheet, hosted by SparkFun**

[www.sparkfun.com/datasheets/Sensors/Biometric/MQ-7.pdf](http://www.sparkfun.com/datasheets/Sensors/Biometric/MQ-7.pdf)

**S26 MQ6 — Winsen MQ-6 heater resistance at room temperature**

[www.winsen-sensor.com/product/mq-6.html](http://www.winsen-sensor.com/product/mq-6.html)

**S27 MQ2 — Winsen MQ-2 heater resistance at room temperature**

[www.winsen-sensor.com/product/mq-2.html](http://www.winsen-sensor.com/product/mq-2.html)

**S28 INA333 — TI quiescent current, output unloaded**

[www.ti.com/lit/ds/symlink/ina333.pdf](http://www.ti.com/lit/ds/symlink/ina333.pdf)

**S29 MCP4725 — Microchip supply current, DS20002039E p3**

[ww1.microchip.com/downloads/aemDocuments/documents/MSLD/ProductDocuments/DataSheets/MCP4725-Data-Sheet-20002039E.pdf](http://ww1.microchip.com/downloads/aemDocuments/documents/MSLD/ProductDocuments/DataSheets/MCP4725-Data-Sheet-20002039E.pdf)

**S30 TPS22919 — TI switch supply current and on-resistance**

[www.ti.com/lit/ds/symlink/tps22919.pdf](http://www.ti.com/lit/ds/symlink/tps22919.pdf)

**S31 LM321DTR (XBLW) — XBLW original datasheet v1.0 p2; distributor mirror**

[xonstorage.z8.web.core.windows.net/pdf/xblw\\_lm321dtrxblw\\_xonlink.pdf](http://xonstorage.z8.web.core.windows.net/pdf/xblw_lm321dtrxblw_xonlink.pdf)

**S32 CH340C — WCH original datasheet v3D, section 6.3; distributor mirror**

[www.hestore.eu/en/prod\\_getfile.php?id=21171](http://www.hestore.eu/en/prod_getfile.php?id=21171)

Full URLs and calculation inputs: power-sources.json, power-calculations.json and power\_budget.py in the diagram package.